Srikrishnaa **Vadivel** | Computational Physicist | First Principles | HPC | Scientific Software

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Profile

Computational physics PhD researcher focused on first-principles calculations of self-trapped excitons, scientific software, and high-performance simulation. Experience includes leadership-class HPC, PETSc/SLEPc, MPI/OpenMP, Python/C++, and ML-assisted materials workflows.

Research

Yale University
PhD Researcher — *2021–present*

- Perform first-principles calculations of self-trapped excitons using HPC resources at Oak Ridge, Aurora, LBNL, and LLNL.
- Build reproducible Linux, Bash, Python, C/C++, CMake, PETSc/SLEPc, MPI/OpenMP, Docker, and Kubernetes workflows for large simulations.
- Explore machine-learning surrogates and equivariant molecular models for materials and molecular systems.

Selected Technical Projects

DOLFINx PDE Gallery: Weak forms for Poisson, heat, and elasticity, plus solver and visualization scaffold.

First-Principles LiF: Plane-wave pseudopotential Hamiltonian and band plotting.

Physics Action Notes: Path integrals, imaginary time, EM, GR, QED/QCD, Standard Model, and strings from action principles.

Meep Ring Resonator: Photonics FDTD geometry and source setup.

Industry

Probe Technologies / Weatherford
Software and Embedded Engineer — *2019–2021*
Built CUDA/OpenGL waveform processing and visualization software for acoustic logging tools; also delivered embedded sensor logging firmware and FPGA logic.

Education

Yale University
PhD, Electrical Engineering — *2021–present*

Yale University
MPhil, Electrical Engineering — *2023*

Yale University
MS, Electrical Engineering — *2023*

Cornell University
MEng, Electrical and Computer Engineering — *2017–2018*

Cornell University
BA, Electrical and Computer Engineering — *2013–2017*

Skills

Methods: DFT workflows, PDEs, numerical linear algebra
Scientific: PETSc, SLEPc, DOLFINx concepts, Meep concepts
Systems: MPI, OpenMP, Linux, CMake
Programming: Python, C++, CUDA, PyTorch